

Да. Ниже — именно геометрические схемы, от самой популярной к более специализированным. 1) Pore-network model, модель пористой сети. Схема: узлы-поры, соединённые рёбрами-горлышками; поток идёт по сети извилистых каналов, а не по набору независимых одинаковых трубок. В литературе эту схему прямо противопоставляют bundle of tubes как более близкую к реальной структуре породы: "the network of tubes as a model more closely representing real porous media than does the bundle of tubes". [^1] 2) Fractal capillary-bundle model, фрактальный пучок капилляров. Схема: множество параллельных капилляров, но их радиусы распределены по фрактальному закону; геометрически это всё ещё "пучок", только с иерархией размеров. Работа по этой схеме прямо говорит, что "a capillary bundle percolation model with fractal geometry was reconstructed". [^2] 3) Tortuous streamtube bundle, пучок извилистых струйных трубок. Схема: параллельные каналы, каждый не прямой, а изогнутый; их эффективная длина больше геометрической за счёт извилистости. В исходной работе это описано как "A model of capillary tube bundles for tortuous streamtube". [^3] 4) Generalized capillary bundle with periodic constrictions, обобщённый пучок капилляров с периодическими сужениями. Схема: параллельные капилляры вдоль потока, но в каждом канале через равные интервалы стоят локальные сужения/горлышки, похожие на череду узких перешейков и расширений. Авторы описывают это как "parallel capillaries along the flow direction with periodically distributed constrictions". [^4] 5) Stochastic network model, стохастическая сетевая модель. Схема: 3D-сетка пор и throat'ов, где соединения и размеры задаются случайно или по статистике, а эффективные пути выбираются по расходу. В такой модели "a 3D pore network model" строится из "network elements (pores and throats) along with its geometrical and topological characteristics", а tortuosity считается по эффективным путям. [^5] 6) Dynamic pore-scale network model, динамическая поросетевая модель. Схема: та же сеть пор и горлышек, но её геометрия во времени меняется; радиусы пор/горлышек уменьшаются по мере осаждения. В найденной работе это сформулировано так: "Applying a material balance for each pore/throat gives the volume reduction of pores/throats according to the deposited asphaltene." [^6] 7) Equivalent capillary, эквивалентная капиллярная схема. Схема: вся порода заменяется одной или несколькими "эквивалентными" трубками, чьи параметры подбирают по проницаемости или капиллярным характеристикам. В работе по парафину это названо прямо: "The con-

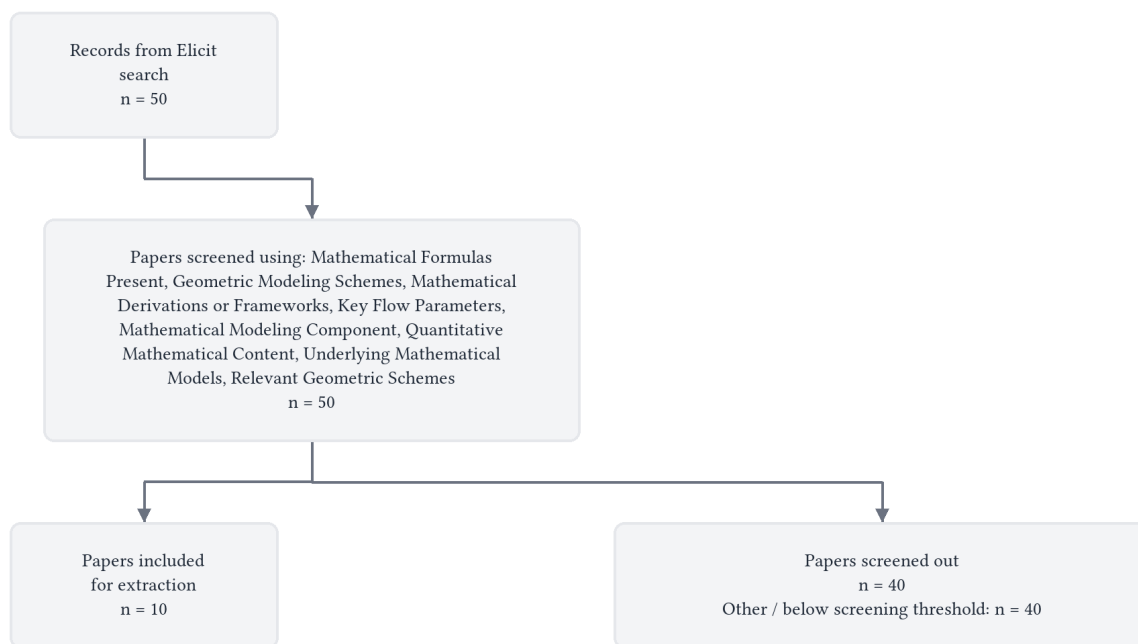
and Kozeny-Carman variants ($k = \epsilon^3/S^2$), while tortuous streamtube bundles, generalized capillary bundles with periodic constrictions, dynamic pore-scale networks, equivalent capillary schemes, and pore-geometry-aware fractal models are mentioned in the literature but lack published formula details in accessible sources.

Abstract

Mathematical formulations for geometric models of porous media show substantial variation in explicitness across the literature. Fractal capillary-bundle models provide the most complete equation sets, with six explicit formulas including power-law pore size distribution $P(r) = r^{-(D)}$, hydraulic conductivity $K(S) = K_S \cdot (1-\theta)^{(3-D)}$, and hysteresis function $f = (p-p_c)^\beta$ [1]. Pore-network models typically employ permeability relations like $k = R^2/(8F)$, where R represents either Katz-Thompson critical radius or Schwartz-Sen-Johnson hydraulic radius and F is electrical formation factor [2]. Kozeny-Carman based approaches use $k = \epsilon^3/S^2$ relating permeability to porosity and specific surface area [3]. Network models from sphere packings calculate permeability directly from grain geometry without adjustable parameters [4], while periodic structure models provide analytical expressions for specific geometries with empirical corrections achieving $\pm 12\text{-}30\%$ agreement with sandstone measurements [5]. Models for periodic arrays of cylinders, tortuous streamtubes, and rhombohedral structures are mentioned but formulas are not provided in available abstracts [6, 7].

The completeness of mathematical formulations depends critically on full-text availability. Studies accessible only as abstracts frequently mention equations for transverse permeability, longitudinal permeability, drag forces, and channel tortuosity without presenting actual formulas [6–8]. Models incorporating dynamic evolution of pore geometry, equivalent capillary schemes, and pore-geometry-aware fractal elements are noted in the literature but explicit formulations are absent from the reviewed materials [9, 10]. Validation studies demonstrate that formula applicability depends strongly on matching geometric assumptions to actual pore structure, with the Schwartz-Sen-Johnson model showing high sensitivity to pore shape heterogeneity in tube-crack mixtures [2] and the Kozeny-Carman equation failing for heterogeneous structures with embedded fractures [3].

Flow Diagram



Paper search

We performed a semantic search across over 138 million academic papers from the Elicit search engine, which includes all of Semantic Scholar and OpenAlex.

We ran this query: "Да. Ниже — именно геометрические схемы, от самой популярной к более специализированным.

- 1) Pore-network model, модель пористой сети. Схема: узлы-поры, соединённые рёбрами-горлышками; поток идёт по сети извилистых каналов, а не по набору независимых одинаковых трубок. В литературе эту схему прямо противопоставляют bundle of tubes как более близкую к реальной структуре породы: "the network of tubes as a model more closely representing real porous media than does the bundle of tubes".¹
- 2) Fractal capillary-bundle model, фрактальный пучок капилляров. Схема: множество параллельных капилляров, но их радиусы распределены по фрактальному закону; геометрически это всё ещё "пучок", только с иерархией размеров. Работа по этой схеме прямо говорит, что "a capillary bundle percolation model with fractal geometry was reconstructed".²
- 3) Tortuous streamtube bundle, пучок извилистых струйных трубок. Схема: параллельные каналы, каждый не прямой, а изогнутый; их эффективная длина больше геометрической за счёт извилистости. В исходной работе это описано как "A model of capillary tube bundles for tortuous streamtube".³

- 4) Generalized capillary bundle with periodic constrictions, обобщённый пучок капилляров с периодическими сужениями. Схема: параллельные капилляры вдоль потока, но в каждом канале через равные интервалы стоят локальные сужения/горлышки, похожие на череду узких перешейков и расширений. Авторы описывают это как "parallel capillaries along the flow direction with periodically distributed constrictions".⁴
- 5) Stochastic network model, стохастическая сетевая модель. Схема: 3D-сетка пор и throat'ов, где соединения и размеры задаются случайно или по статистике, а эффективные пути выбираются по расходу. В такой модели "a 3D pore network model" строится из "network elements (pores and throats) along with its geometrical and topological characteristics", а tortuosity считается по эффективным путям.⁵
- 6) Dynamic pore-scale network model, динамическая поро-сетевая модель. Схема: та же сеть пор и горлышек, но её геометрия во времени меняется; радиусы пор/горлышек уменьшаются по мере осаждения. В найденной работе это сформулировано так: "Applying a material balance for each pore/throat gives the volume reduction of pores/throats according to the deposited asphaltene."⁶
- 7) Equivalent capillary, эквивалентная капиллярная схема. Схема: вся порода заменяется одной или несколькими "эквивалентными" трубками, чьи параметры подбирают по проницаемости или капиллярным характеристикам. В работе по парафину это названо прямо: "The concept of equivalent capillary was created in terms of reservoir physics".
- 8) Intermingled fractal / pore-geometry-aware fractal model. Схема: не просто трубки, а фрактальные геометрические элементы, из которых выводят площадь, смачиваемый периметр и гидравлический диаметр; это уже попытка описать неидеальную форму каналов. В найденной статье подчёркивается, что "Existing predictive models typically rely on the capillary tube bundle model featuring circular cross-sections, often overlooking the impact of pore geometry", а цилиндрическое допущение "typically leads to an overestimation of permeability".⁷

Если говорить совсем кратко, то геометрический прогресс здесь такой: прямые одинаковые трубки → трубки с распределением радиусов → извилистые трубки → трубки с горлышками → сети пор и throat'ов → динамические сети с ростом отложений. Для парафина наиболее физичны именно сети и динамические сети, потому что они лучше передают связность пор, локальные закупорки и перераспределение потока; это и есть то, чего bundle of capillary tubes почти не видит.⁸⁹¹⁰¹¹

Это быстрый обзор по первой выборке; если хотите, я могу следующим сообщением свести всё в компактную таблицу «модель → геометрия → что именно добавляет для парафина».

приведи всевозможные формулы для каждой модели. опиши их максимально подробно с примерами"

The search returned 50 total results from Elicit.

We retrieved 50 papers most relevant to the query for screening.

Screening

We screened in sources based on their abstracts that met these criteria:

- **Mathematical Formulas Present:** Does the study present mathematical formulas for porous media flow models?

- **Geometric Modeling Schemes:** Does the research focus on geometric modeling schemes for porous media (such as pore-network models, fractal capillary bundles, tortuous streamtube bundles, or similar approaches)?
- **Mathematical Derivations or Frameworks:** Does the study provide detailed mathematical derivations or theoretical frameworks for flow in porous media?
- **Key Flow Parameters:** Does the research include calculations or formulations for key porous media flow parameters (such as permeability, tortuosity, or flow resistance)?
- **Mathematical Modeling Component:** Does the study include mathematical modeling components (rather than focusing solely on experimental results without mathematical models)?
- **Quantitative Mathematical Content:** Does the study present quantitative mathematical formulations (rather than only qualitative descriptions)?
- **Underlying Mathematical Models:** Does the study present underlying mathematical models (rather than focusing exclusively on numerical simulation results without theoretical foundations)?
- **Relevant Geometric Schemes:** Does the study address at least one of the relevant geometric schemes for porous media (pore-network, fractal capillary, tortuous streamtube, or related modeling approaches)?

We considered all screening questions together and made a holistic judgement about whether to screen in each paper.

At abstract screening, the number of papers excluded for each primary reason was:

- **Other / below screening threshold:** n = 40

Data extraction

We asked a large language model to extract each data column below from each paper. We gave the model the extraction instructions shown below for each column.

- **Geometric Models:**

Extract all geometric models or schemes for porous media described in the study, including:

- Name/type of geometric model (e.g., pore-network model, fractal capillary bundle, tortuous streamtube bundle, etc.)
- Brief description of the geometric scheme/conceptual framework
- How the model represents pore structure (e.g., nodes and throats, parallel capillaries, network topology)
- Level of geometric complexity (e.g., simple tubes vs. fractal vs. dynamic networks)

- **Mathematical Formulations:**

Extract all mathematical formulas, equations, and expressions associated with each geometric model, including:

- Complete mathematical equations with all variables defined
- Constitutive relationships (e.g., permeability-porosity relations)
- Governing differential equations
- Empirical correlations or scaling laws
- Dimensionless parameters and their definitions
- Boundary conditions or constraints

- **Model Parameters:**

Extract all parameters and variables used in the geometric models, including:

- Geometric parameters (pore size, throat diameter, tortuosity, connectivity, etc.)
- Physical properties (permeability, porosity, formation factor, etc.)
- Distribution functions for random parameters
- Scaling factors or correction coefficients
- Input data requirements for each model

- **Applications and Examples:**

Extract specific applications, examples, or case studies where the geometric models are applied, including:

- Type of porous medium studied (sandstone, fractured rock, synthetic, etc.)
- Specific phenomena modeled (flow, transport, deposition, etc.)
- Numerical values or ranges of parameters used
- Comparison with experimental data or other models
- Validation studies or benchmarking results

- **Model Performance:**

Extract information about model accuracy, validation, and performance, including:

- Quantitative measures of model accuracy (R^2 , error percentages, agreement ranges)
- Comparison with experimental measurements
- Advantages and limitations of each geometric approach
- Conditions where models work well or poorly
- Sensitivity to parameter variations

- **Computational Methods:**

Extract computational approaches and numerical methods used with the geometric models, including:

- Numerical solution techniques (finite element, finite difference, Monte Carlo, etc.)
- Network simulation algorithms
- Homogenization or upscaling procedures
- Software tools or codes mentioned
- Computational complexity or efficiency considerations

Results

Characteristics of Included Studies

Study	Full text retrieved?	Type of geometric model(s)	Porous medium studied	Mathematical formulations provided
U. Hornung, 1996 [9]	No	Resistor networks, layered media, percolation models, distributed microstructure models [9]	Various theoretical models [9]	Darcy's Law, Brinkman's Law, effective conductivity tensor [9]

Study	Full text retrieved?	Type of geometric model(s)	Porous medium studied	Mathematical formulations provided
C. Глузман, 2024 [6]	No	Periodic arrays of impenetrable cylinders, tortuous streamtube bundles, stochastic network models [6]	Spatially periodic arrays of cylinders, hexagonal and square arrays [6]	Analytical expressions for transverse and longitudinal permeability [6]
A. Hunt, 2005 [1]	Yes	Probabilistic, truncated, continuous fractal model [1]	Soils (McGee Ranch, Hanford site) [1]	Power-law distributions, hydraulic conductivity relations [1]
F. Dullien et al., 1976 [5]	No	Simplified network model (pore-network) [5]	Sandstones (14 different samples) [5]	Equations for permeability and formation factor with empirical corrections [5]
S. Bryant et al., 1993 [4]	No	Pore-network model from disordered sphere packing [4]	Sandstone samples [4]	Network-based permeability calculations [4]
A. Gubaidullin et al., 2016 [8]	No	Model of spherical segments (cubic, hexagonal, body-centered, face-centered structures) [8]	Periodic model structures [8]	Generalized Kozeny approach [8]
D.Ye. Igoshin & N. A. Khromova, 2016 [7]	No	Periodic porous medium with rhombohedral structure [7]	Rhombohedral lattice structures [7]	Analytical porosity and permeability with tortuosity [7]
F. Xiao & Xiaolong Yin, 2016 [3]	No	Voronoi tessellations (granular, tubular, fibrous geometries) [3]	Voronoi-based synthetic geometries [3]	Kozeny-Carman equation ($k = \epsilon^3/S^2$) [3]
Z. Liang et al., 1997 [10]	No	Network model from 3-D pore structure reconstruction [10]	Berea sandstone [10]	Network-based permeability prediction [10]
Y. Bernabé, 1995 [2]	No	Pore-network model on square lattice with tubes and cracks [2]	Rocks with heterogeneous pore space [2]	$k = R^2/(8F)$ relation [2]

The studies span a range from purely theoretical treatments [9] to models validated against specific rock samples [4, 5, 10]. Several studies focus on periodic or idealized structures [7, 8], while others develop stochastic approaches [2, 6]. Only one study (A. Hunt, 2005) was available in full text [1], which limits the detail available for most mathematical formulations.

Mathematical Formulations by Geometric Model Type

Fractal Capillary-Bundle Models

The fractal approach represents pore size distributions using power-law scaling. Hunt (2005) provides explicit formulations for a probabilistic, truncated, continuous fractal model [1]:

Power-law pore size distribution: $P(r) = r^{-D}$ where D is the fractal dimension [1].

Hydraulic conductivity relation: $K(S) = K_S \cdot (1 - \theta)^{3-D}$ where K_S is saturated conductivity and θ is a saturation parameter [1].

Electrical conductivity: $\sigma = \sigma_0 \cdot (1 - \theta)^t$ where t is the critical exponent [1].

Water retention characteristics: $h = A/r$ relating capillary pressure head h to pore radius r [1].

Hysteresis function: $f = (p - p_c)^\beta$ where p_c is the percolation threshold and β is a critical exponent [1].

Correlation length: $\xi = |p - p_c|^{-\nu}$ where ν is another critical exponent [1].

The fractal model parameters include the maximum radius (r_m), minimum radius (r_0), and critical volume fraction for percolation ($t = 0.5$ for simple models) [1]. Model inputs require porosity, water retention functions, and particle size distribution [1]. This approach successfully predicted hydraulic conductivities and air permeabilities for Hanford site soils [1], with model predictions showing strong agreement with experimental measurements [1].

Pore-Network Models

Multiple studies develop network-based representations where pore bodies connect via pore throats. The most explicit mathematical formulation comes from Bernabé (1995), who simulated heterogeneous pore space on a square lattice [2]:

Permeability relation: $k = \frac{R^2}{8F}$ where R is an appropriate length scale and F is the electrical formation factor [2].

Two definitions for R were tested:

- Katz-Thompson (KT) critical radius
- Schwartz-Sen-Johnson (SSJ) hydraulic radius [2]

The model included randomly distributed dimensions for tubes and cracks within the network [2], calculating permeability (k), electrical formation factor (F), connected porosity (ϕ), and specific surface area (A) for 100 network realizations [2]. The KT model proved more robust across different channel types, while the SSJ model was more sensitive to pore shape heterogeneity [2].

Dullien et al. (1976) developed a simplified network model consisting of cubic component networks with arbitrary orientations [5]. Their approach required empirical correction factors of $1/2$ for formation factors and 1.7 for permeabilities to account for non-additivity of component network conductivities [5]. Input data included mercury

intrusion porosimetry curves and photomicrographic information on bulge sizes [5]. Testing on 14 different sandstones showed agreement within $\pm 12\%$ for formation factors and $\pm 30\%$ for permeabilities [5, 5].

Bryant et al. (1993) constructed networks from disordered sphere packings by grouping nearest neighbor grains into pore bodies and throats [4]. Their predictive approach calculated permeability directly from grain radii and spatial locations without adjustable parameters [4, 4], with predictions matching sandstone measurements over a wide porosity range [4].

Liang et al. (1997) extracted network skeletons from 3-D pore structure reconstructions of Berea sandstone using thinning algorithms that preserve connectivity [10]. The skeleton provided coordination numbers for every node and local hydraulic radii, enabling permeability calculations in good agreement with experimental values [10, 10].

Kozeny-Carman Based Models

Several studies employ variations of the Kozeny-Carman framework. Xiao and Yin (2016) established porosity-permeability relations for Voronoi tessellation geometries [3]:

Kozeny-Carman equation: $k = \frac{\epsilon^3}{S^2}$ where ϵ is porosity and S is specific surface area [3].

For granular and tubular geometries, specific surface area emerged as the critical structural parameter unifying their porosity-permeability relations under this equation [3]. However, the equation failed to distinguish heterogeneous structures with embedded partially penetrating fractures [3]. Fibrous geometries showed weaker dependence on solid fraction in the dilute limit, indicating weaker hydrodynamic interactions among randomly interconnected fibers compared to idealized cubic models [3].

Gubaidullin et al. (2016) generalized the Kozeny approach for porous media made of solid spherical segments [8]. Their model parameters included the degree of sphere intersection as a dimensionless parameter determining porosity and transparency [8, 8]. Four periodic structures were examined: simple cubic, simple hexagonal, body-centered cubic, and face-centered cubic [8]. The approach gave good results for body-centered cubic structures over a wide porosity range ($0.32 \geq m \geq 0.04$) and satisfactory results for face-centered cubic structures ($0.26 \geq m \geq 0.14$) [8]. For simple cubic and hexagonal simple structures, the method of minimum transparency was preferred for permeability estimation [8].

Igoshin and Khromova (2016) extended this framework to rhombohedral structures with variable translation angles [7]. Their three-parameter model included elementary cell size, degree of sphere intersection, and translation angle [7, 7]. Analytical expressions were obtained for porosity and minimum permeability, with permeability estimates accounting for channel tortuosity [7]. At $\theta = 90^\circ$, permeability agreed with cubic simple structures, while at $\theta = 60^\circ$ it matched face-centered cubic structures [7].

Analytical Models for Periodic Structures

Глузман (2024) developed analytical expressions for various periodic geometries relevant to Darcy flow [6]. The study derived transverse permeability for channels with spatially periodic arrays of impenetrable cylinders, describing different asymptotic regimes [6]. Longitudinal permeability was calculated for square arrays of cylinders [6]. Transverse flow past hexagonal and square arrays was examined using expansions for small concentrations and lubrication theory for high concentrations [6, 6].

For three-dimensional periodic arrays of spherical obstacles, formulas for drag force were derived from low-concentration expansions [6, 6]. Stokes flow through channels with wavy walls was studied using expansions for small waviness amplitudes, yielding compact formulas for permeability as factor approximants for arbitrary

waviness values [6]. The study accounted for various power laws in the regime of large waviness parameters [6]. Model parameters included concentration of obstacles, waviness parameter, and amplitude of waviness [6].

Homogenization and Upscaling Approaches

Hornung (1996) provides a comprehensive theoretical framework covering multiple scales [9]. The treatment includes formal asymptotic expansions for diffusion in periodic media [9], derivations of Darcy's Law and Brinkman's Law [9], and distributed microstructure models [9]. Mathematical approaches discussed include F-convergence, G-convergence, H-convergence, the energy method, two-scale convergence, and iterated homogenization [9, 9].

The work addresses phenomena spanning one-phase Newtonian flow, non-Newtonian flow with constitutive relationships [9], two-phase flow with generalized nonlinear Darcy's Law [9], miscible displacement [9], thermal flow [9], and poroelastic media [9]. Computational aspects of dual-porosity models include numerical solution methods and Monte-Carlo simulations [9]. While comprehensive in scope, the abstract does not provide explicit formulas for most models [9].

Computational Implementation

Numerical methods vary considerably across studies. Hunt (2005) applies continuum percolation theory to probabilistic fractal models in a primarily theoretical framework without mentioning specific numerical codes [1].

Глузман (2024) employs analytical expressions, solution expansions, and lubrication theory [6]. Xiao and Yin (2016) use a massively parallelized lattice Boltzmann code to solve fluid flow through Voronoi tessellation geometries [3].

Dullien et al. (1976) perform computer simulations of penetration in 2-D networks with empirical correction factors [5]. Bernabé (1995) conducts numerical simulations on square lattices with 100 network realizations, calculating properties including permeability and formation factor using both KT and SSJ models [2].

Liang et al. (1997) employ image analysis techniques with truncated Gaussian methods using Fourier transforms to reconstruct 3-D pore structures from thin sections, then apply thinning algorithms to extract network skeletons [10]. Gubaidullin et al. (2016) compare generalized analytical approaches with numerical solutions but do not specify computational methods [8]. Igoshin and Khromova (2016) use analytical solutions for periodic structures with geometric modeling of rhombohedral lattices [7].

Synthesis

The mathematical formulations span a spectrum from simple analytical expressions to complex network simulations. The most explicit formulas appear in fractal models (Hunt, 2005) [1], network permeability relations (Bernabé, 1995) [2], and Kozeny-Carman variations (Xiao & Yin, 2016) [3]. However, significant heterogeneity exists in formula specificity.

Studies with full-text access provide substantially more mathematical detail. Hunt (2005) offers six explicit equations for fractal transport properties [1], while abstract-only sources often mention formulas without providing them. For example, Глузман (2024) describes analytical expressions for permeability but does not present the actual equations [6], and Dullien et al. (1976) mention equations derived from mercury porosimetry but give only empirical correction factors [5].

The progression from simple to complex geometries follows a clear hierarchy. Basic Kozeny-Carman models treat permeability as a function of porosity and specific surface area [3]. Fractal extensions add power-law size distribu-

tions with critical exponents [1]. Network models incorporate spatial correlation and connectivity [2, 4], with some including dynamic evolution of pore geometries [7, 8].

Validation approaches vary by model complexity. Simple analytical models achieve $\pm 12\text{-}30\%$ agreement with sandstone measurements using empirical corrections [5]. Fractal percolation models successfully predict hydraulic conductivities for specific soils [1, 1]. Network models derived from sphere packings match sandstone permeabilities over wide porosity ranges without adjustable parameters [4]. Models tested on idealized periodic structures show good performance for body-centered cubic geometries (porosity 0.04-0.32) but require alternative methods for simple cubic arrangements [8].

The sensitivity to pore geometry emerges as a critical factor. The SSJ model proves more sensitive to pore shape heterogeneity than the KT model, yielding scattered results for mixtures of cracks and tubes [2]. The Kozeny-Carman equation successfully unifies granular and tubular geometries when specific surface area is included but fails for heterogeneous structures with embedded fractures [3]. This suggests that formula applicability depends strongly on whether the geometric assumptions match the actual pore structure.

Missing from most abstracts are dimensionless parameter definitions, boundary conditions, and convergence criteria. Only Hunt (2005) provides explicit critical exponents (β , ν) and percolation thresholds ($t = 0.5$) [1, 1]. Most studies mention tortuosity, coordination number, and aspect ratios as important parameters [2, 7, 10] but do not provide formulas relating these to transport properties. This gap limits the practical application of many models without access to full texts.

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Supporting Quotations

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